

Two-Stage Bidirectional Inverter Equivalent Circuit Model for Distribution Grid Steady-State Analysis and Optimization

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Play With Us!



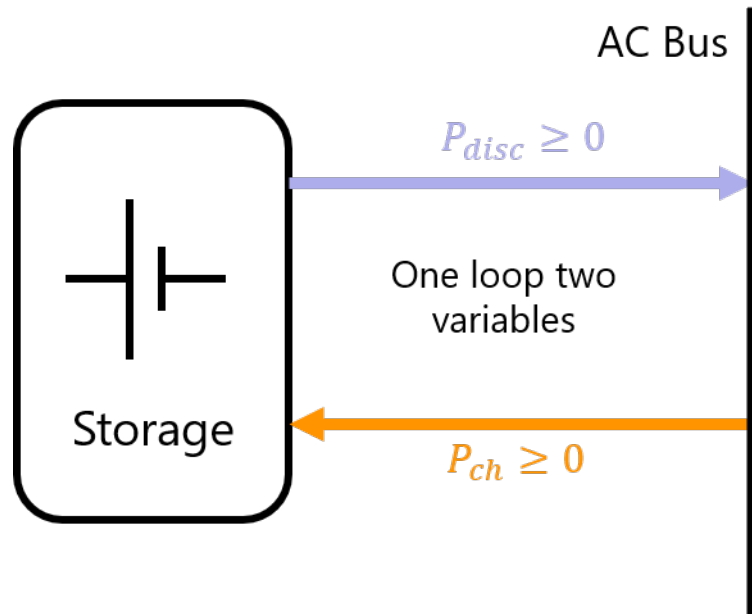
Demo Objective

- Introduce a two-stage bidirectional inverter equivalent circuit model
- Embed the model within a standard BMOPF .json file
- Show a steady-state and optimization study for networks up to 4k+ three-phase (~12k) nodes with 1000s of these inverter models

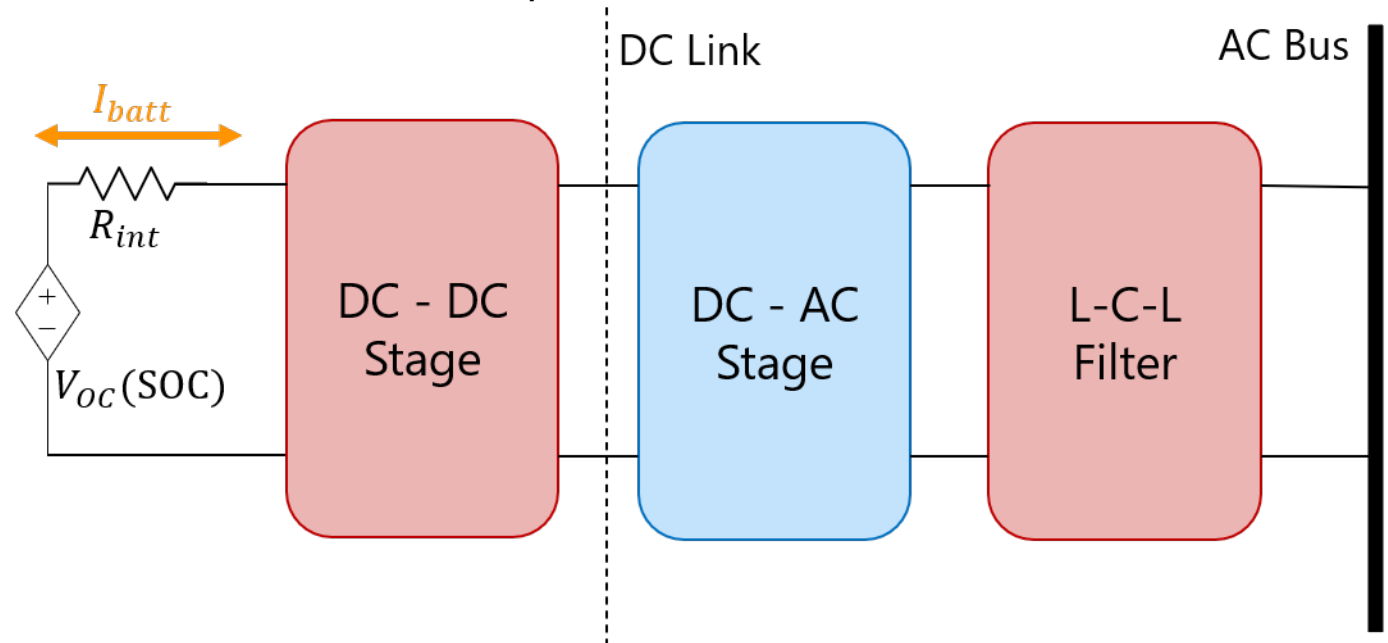
Why **Another** Inverter Model

- Complementarity constraint in a battery is not a physical phenomenon
 - Charge and discharge cannot happen concurrently if there is only one electrical loop

Two power variable model



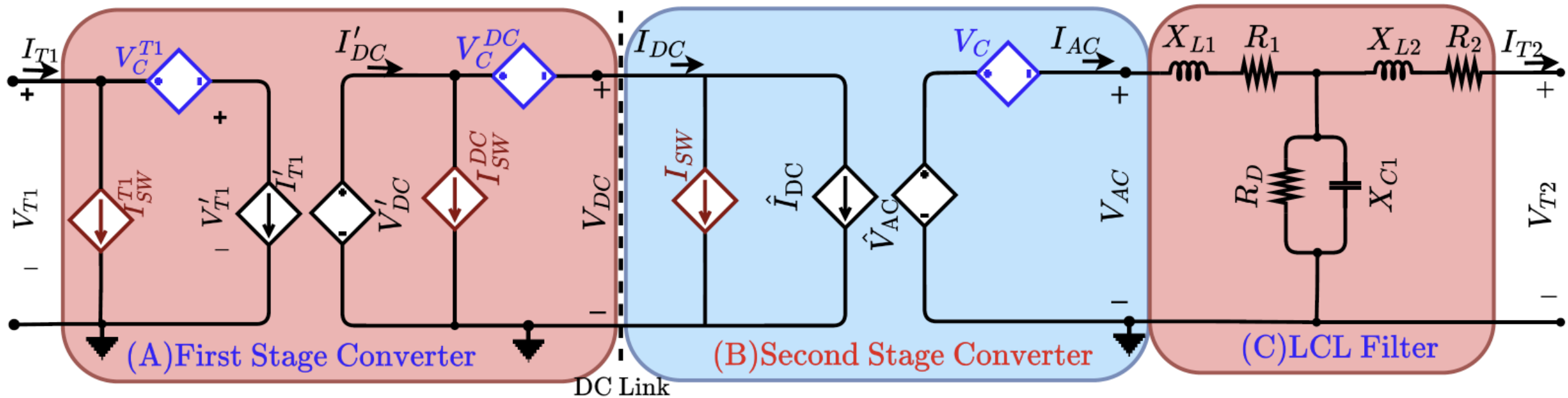
Equivalent circuit model



Steady-state Equivalent Circuit Inverter Model

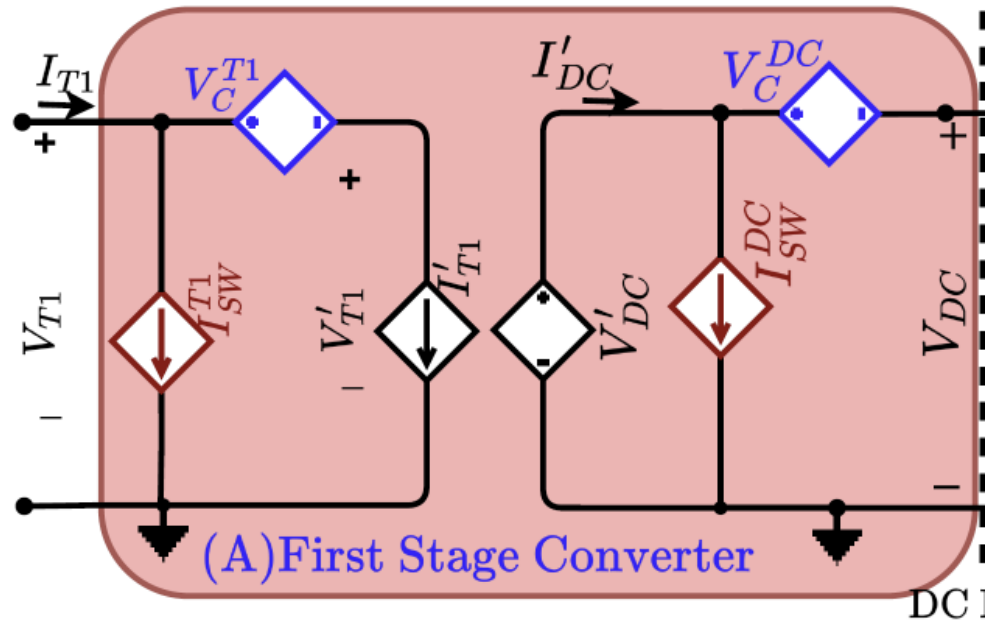
- Physics-based steady-state equivalent circuit model
- Integrates the passive and active components in a unified circuit representation
- Explicitly captures converter and filter losses under bidirectional power flow

Model-at-a-Glance



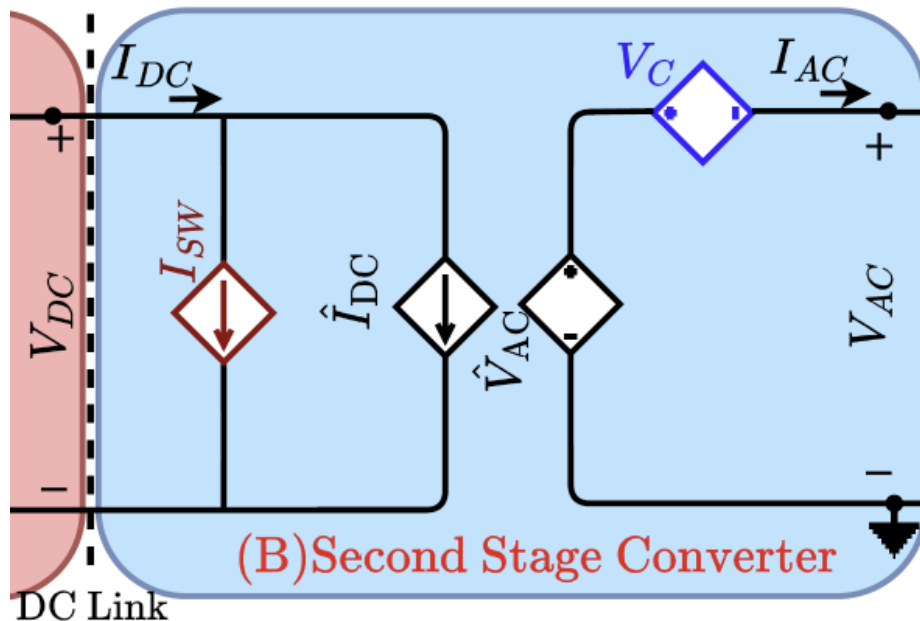
First Stage Converter

- Bidirectional, non-inverting buck–boost DC–DC stage
- Regulates the voltage from different DC sources to a common DC-link voltage



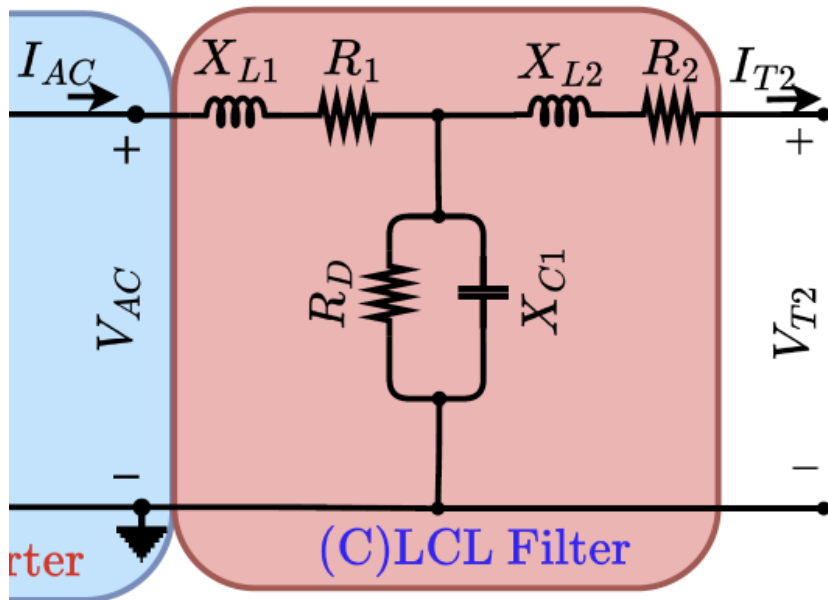
Second Stage Converter

- Bidirectional, converts power between the DC link and AC side
- Uses modulation to regulate the AC-side voltage magnitude and phase



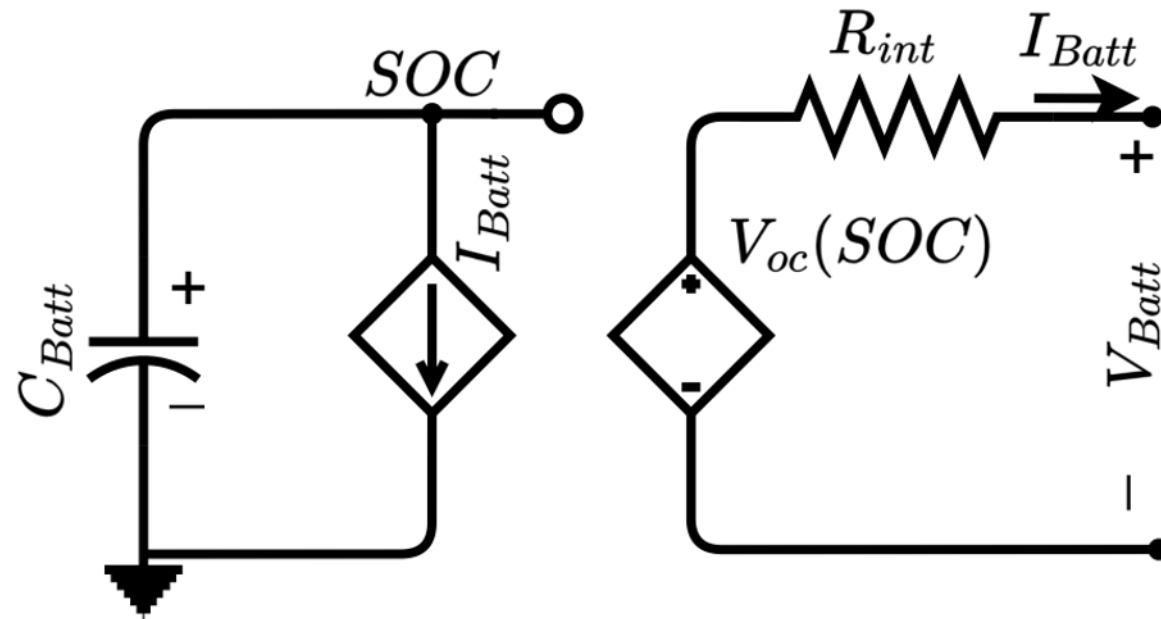
LCL Filter

- Damped inductive-capacitive-inductive AC output filter
- Attenuates high-frequency switching harmonics at the AC-side interface



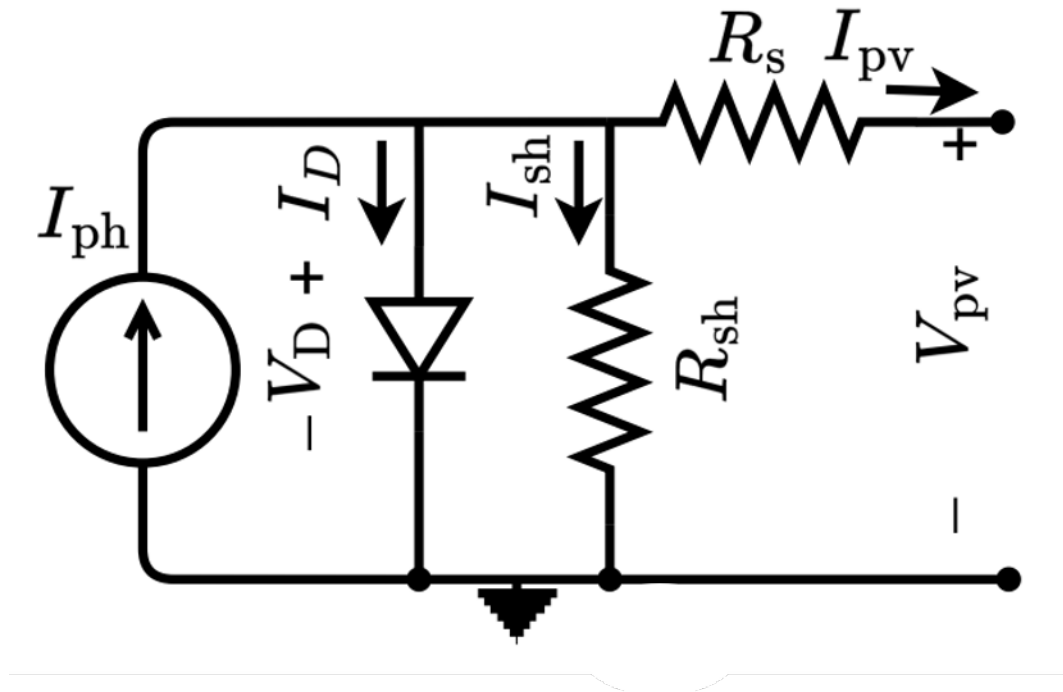
DC Sources – Battery energy storage (BES)

- Zeroth-order equivalent circuit model
- Captures SOC-dependent open-circuit voltage and internal resistance



DC Sources – Photovoltaic (PV) Module

- Single-diode model (SDM) of the PV module
- Captures the nonlinear current–voltage (I-V) characteristics of the PV source



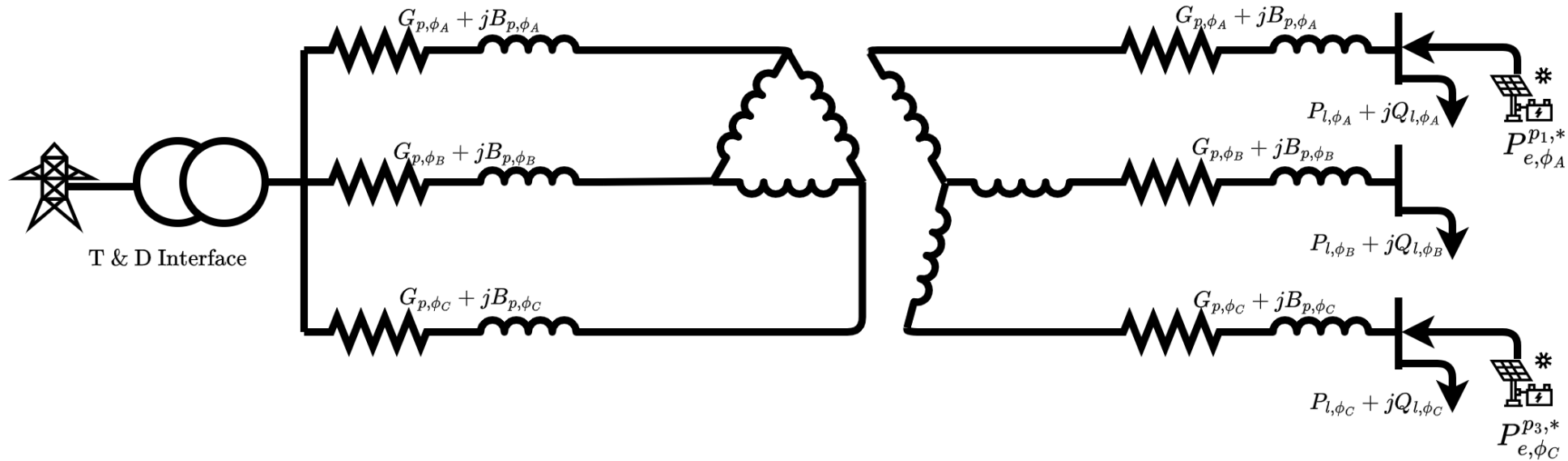
Control Laws

- Active power control
 - Constant active power
 - Maximum power point transfer

Control Laws

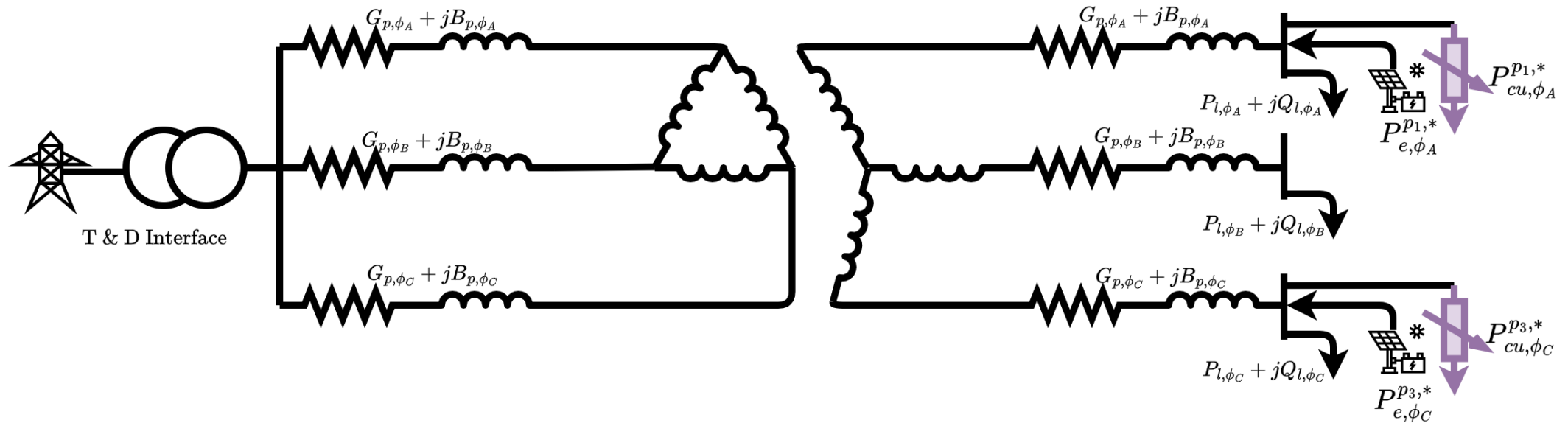
- Active power control
 - Constant active power (CONSTANT_P)
 - Maximum power point transfer (MPPT)
- Reactive power control
 - Constant reactive power
 - Unity power factor
 - Constant power factor
 - Volt-var control

Case Study: Optimal PV Export Curtailment



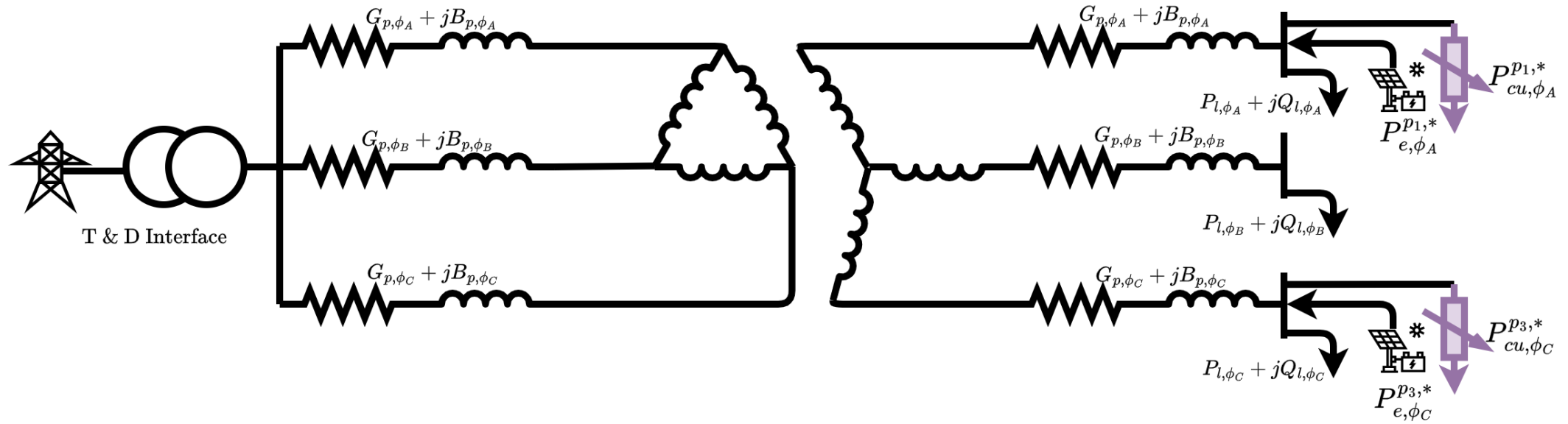
- Baseline: Power export to the grid from the prosumers is possible
- Problem: Too much export may violate the grid and component operating limits

Case Study: Optimal PV Export Curtailment



- Introduce a **curtailment power object** for each PV object
- The goal is to meet grid operating criteria while minimizing **curtailment** across all PVs

Case Study: Optimal PV Export Curtailment



$$\min \quad \left\| \mathbf{P}_{cu} \right\|_p, \quad p \in \{1, 2, \infty\}$$

s.t. AC network constraints

Flow and nodal voltage bounds

Curtailment constraints

Demo time!

